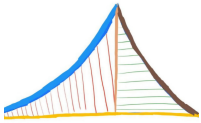
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE MATHEMATICS 2017

A. Formulae

1.1.1 A tangent to a circle with centre $\mathbf{O}$ is perpendicular to the radius through the point of contact $\mathbf{T}$,

$$(\mathbf{P}-\mathbf{T})^{\top}(\mathbf{T}-\mathbf{O})=0 \tag{1.1.1.1}$$

where $\mathbf{P}$ is any other point on the tangent.

1.1.2 The lengths of the two tangents drawn from an external point $\mathbf{P}$ to a circle of radius a and centre $\mathbf{O}$ are equal and

$$\|\mathbf{P}-\mathbf{T}_1\|=\|\mathbf{P}-\mathbf{T}_2\|=\sqrt{\|\mathbf{P}-\mathbf{O}\|^2-a^2} \tag{1.1.2.1}$$

where $\mathbf{T}_1$ and $\mathbf{T}_2$ are the points of contact.

Solution: From (1.1.1.1) and Pythagoras' theorem in the right triangle $\mathbf{OT}_i\mathbf{P}$, $i=1,2$,

$$\|\mathbf{P}-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+\|\mathbf{T}_i-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+a^2 \tag{1.1.2.2}$$

which is the same for $i=1,2$, yielding (1.1.2.1).

1.1.3 In a right triangle with an acute angle θ,

$$\sin\theta=\frac{\text{opposite}}{\text{hypotenuse}},\quad \tan\theta=\frac{\text{opposite}}{\text{adjacent}} \tag{1.1.3.1}$$

1.1.4 For complementary angles,

$$\tan\left(90^\circ-\theta\right)=\frac{1}{\tan\theta} \tag{1.1.4.1}$$

1.1.5 The locus of a point $\mathbf{x}$ equidistant from the points $\mathbf{A}$ and $\mathbf{B}$ is

$$(\mathbf{A}-\mathbf{B})^{\top}\mathbf{x}=\frac{\|\mathbf{A}\|^2-\|\mathbf{B}\|^2}{2} \tag{1.1.5.1}$$

Solution:

$$\|\mathbf{x}-\mathbf{A}\|^2=\|\mathbf{x}-\mathbf{B}\|^2 \tag{1.1.5.2}$$

$$\implies \|\mathbf{x}\|^2-2\mathbf{A}^{\top}\mathbf{x}+\|\mathbf{A}\|^2=\|\mathbf{x}\|^2-2\mathbf{B}^{\top}\mathbf{x}+\|\mathbf{B}\|^2 \tag{1.1.5.3}$$

which simplifies to (1.1.5.1).

1.1.6 For the quadratic equation $px^2+qx+r=0$ with roots α and β,

$$\alpha+\beta=-\frac{q}{p},\quad \alpha\beta=\frac{r}{p} \tag{1.1.6.1}$$

1.1.7 The nth term of an A.P. with first term a and common difference d is

$$a_n=a+(n-1)\,d \tag{1.1.7.1}$$

1.1.8 The area of a circle of radius r and of a semicircle of radius r are

$$A_{\text{circle}}=\pi r^2,\quad A_{\text{semicircle}}=\frac{1}{2}\pi r^2 \tag{1.1.8.1}$$

1.1.9 The point $\mathbf{R}$ dividing the line segment joining $\mathbf{P}$ and $\mathbf{Q}$ in the ratio $k:1$ is

$$\mathbf{R} = \frac{k\mathbf{Q} + \mathbf{P}}{k+1} \tag{1.1.9.1}$$

1.1.10 The points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are collinear if and only if

$$\operatorname{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix} = 1 \tag{1.1.10.1}$$

which, for points in a plane, is equivalent to

$$\left|\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}\right| = 0 \tag{1.1.10.2}$$

1.1.11 The roots of the quadratic equation $px^2 + qx + r = 0$ are

$$x = \frac{-q \pm \sqrt{q^2 - 4pr}}{2p} \tag{1.1.11.1}$$

The roots are equal if and only if

$$q^2 - 4pr = 0 \tag{1.1.11.2}$$

1.1.12 The sum of the first n terms of an A.P. with first term a and common difference d is

$$S_n = \frac{n}{2}\left[2a + (n-1)\,d\right] \tag{1.1.12.1}$$

1.1.13 The volume of a cuboid of length l, breadth b and height h is lbh. Water flowing through a canal of cross-sectional area A with speed v for a time t has volume

$$V = A\,v\,t \tag{1.1.13.1}$$

1.1.14 The volume of a hollow cylinder of outer radius R, inner radius r and length h is

$$V = \pi\left(R^2 - r^2\right)h \tag{1.1.14.1}$$

1.1.15 The area of a sector of angle θ of a circle of radius r and the area of a ring between circles of radii R and r are

$$A_{\text{sector}} = \frac{\theta}{360^\circ}\pi r^2, \quad A_{\text{ring}} = \pi\left(R^2 - r^2\right) \tag{1.1.15.1}$$

1.1.16 For a cone of radius r and height h, the slant height l and the curved surface area are

$$l = \sqrt{h^2 + r^2}, \quad A_{\text{cone}} = \pi r l \tag{1.1.16.1}$$

The curved surface area of a hemisphere of radius r is

$$A_{\text{hemisphere}} = 2\pi r^2 \tag{1.1.16.2}$$

1.1.17 For an event E with m favourable outcomes out of N equally likely outcomes, and its complement E',

$$\Pr(E) = \frac{m}{N}, \quad \Pr(E) + \Pr(E') = 1 \tag{1.1.17.1}$$

1.1.18 The volume of a cylinder of radius r and height h is

$$V = \pi r^2 h \tag{1.1.18.1}$$

1.1.19 The mth term of an A.P. is related to the sum of its first $2m-1$ terms by

$$S_{2m-1} = (2m-1)\, a_m \tag{1.1.19.1}$$

Solution: From (1.1.12.1), with $n = 2m-1$,

$$S_{2m-1} = \frac{2m-1}{2}\left[2a + (2m-2)\, d\right] = (2m-1)\left(a + (m-1)\, d\right) \tag{1.1.19.2}$$

yielding (1.1.19.1) from (1.1.7.1).

1.1.20 If a person completes a work in t days, the rate of work is $\frac{1}{t}$ of the work per day. The rates of persons working together add up,

$$\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \tag{1.1.20.1}$$

1.1.21 The angle of depression of an object from a point at a height h is equal to the angle of elevation of that point from the object (alternate angles). If the horizontal distance is d, from (1.1.3.1),

$$\tan\theta = \frac{h}{d} \tag{1.1.21.1}$$

1.1.22 An angle in a semicircle is a right angle. If BC is a diameter of a circle and A is a point on the circle, then

$$\angle BAC = 90^\circ, \quad BC^2 = AB^2 + AC^2 \tag{1.1.22.1}$$

The area of a right triangle with perpendicular sides p and q is $\frac{1}{2}pq$.

B. Problems

1.2.1 If the angle between two tangents drawn from an external point P to a circle of radius a and centre O, is 60°, then find the length of OP. (CBSE 2017)

Solution: Let $\mathbf{T}_1$ and $\mathbf{T}_2$ be the points of contact of the tangents from $\mathbf{P}$. From (1.1.2.1),

$$\|\mathbf{P} - \mathbf{T}_1\| = \|\mathbf{P} - \mathbf{T}_2\| \tag{1.2.1.1}$$

Also $\|\mathbf{O} - \mathbf{T}_1\| = \|\mathbf{O} - \mathbf{T}_2\| = a$ and $\mathbf{OP}$ is common. So the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are congruent, and $\mathbf{OP}$ bisects the angle between the tangents,

$$\angle \mathbf{OPT}_1 = \frac{60^\circ}{2} = 30^\circ \tag{1.2.1.2}$$

From (1.1.1.1), the triangle $\mathbf{OT}_1\mathbf{P}$ is right angled at $\mathbf{T}_1$. Using (1.1.3.1),

$$\sin 30^\circ = \frac{a}{\|\mathbf{O} - \mathbf{P}\|} \tag{1.2.1.3}$$

$$\implies \|\mathbf{O} - \mathbf{P}\| = \frac{a}{\frac{1}{2}} = 2a \tag{1.2.1.4}$$

1.2.2 A circle touches all the four sides of a quadrilateral ABCD. Prove that

$$AB + CD = BC + DA$$

(CBSE 2017)

Solution: Let the circle touch AB, BC, CD and DA at **P**, **Q**, **R** and **S** respectively. The tangents drawn from each of **A**, **B**, **C** and **D** are equal, from (1.1.2.1). Hence,

$$AP = AS, \quad BP = BQ, \quad CR = CQ, \quad DR = DS \tag{1.2.2.1}$$

Adding the above,

$$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \tag{1.2.2.2}$$

Since **P** lies on AB, **R** on CD, **S** on DA and **Q** on BC,

$$AP + BP = AB, \quad CR + DR = CD, \quad AS + DS = DA, \quad BQ + CQ = BC \tag{1.2.2.3}$$

Substituting in the above,

$$AB + CD = BC + DA \tag{1.2.2.4}$$

1.2.3 If the distances of $\mathbf{P}\,(x, y)$ from $\mathbf{A}\,(5, 1)$ and $\mathbf{B}\,(-1, 5)$ are equal, then prove that $3x = 2y$. (CBSE 2017)

Solution: Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \tag{1.2.3.1}$$

Then

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}, \quad \|\mathbf{A}\|^2 = 26, \quad \|\mathbf{B}\|^2 = 26 \tag{1.2.3.2}$$

Substituting in (1.1.5.1),

$$\begin{pmatrix} 6 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{26 - 26}{2} = 0 \tag{1.2.3.3}$$

$$\implies 6x - 4y = 0 \tag{1.2.3.4}$$

$$\therefore 3x = 2y \tag{1.2.3.5}$$

1.2.4 Find the value of p, for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other. (CBSE 2017)

Solution: Let the roots be α and 6α. From (1.1.6.1),

$$\alpha + 6\alpha = \frac{14}{p} \tag{1.2.4.1}$$

$$\alpha\,(6\alpha) = \frac{8}{p} \tag{1.2.4.2}$$

From (1.2.4.1),

$$\alpha = \frac{2}{p} \tag{1.2.4.3}$$

Substituting in (1.2.4.2),

$$\frac{24}{p^2} = \frac{8}{p} \tag{1.2.4.4}$$

$$\implies p = 3 \tag{1.2.4.5}$$

For $p = 3$, the roots are 4 and $\frac{2}{3}$, and indeed $4 = 6\left(\frac{2}{3}\right)$. The following code verifies the solution

codes/quadratic.c

1.2.5 For what value of n, are the nth terms of two A.Ps $63, 65, 67, \ldots$ and $3, 10, 17, \ldots$ equal ? (CBSE 2017)

Solution: For the first A.P., $a = 63$, $d = 2$ and for the second A.P., $a = 3$, $d = 7$. Equating the nth terms using (1.1.7.1),

$$63 + (n-1)\,2 = 3 + (n-1)\,7 \tag{1.2.5.1}$$

$$\implies 5\,(n-1) = 60 \tag{1.2.5.2}$$

$$\therefore n = 13 \tag{1.2.5.3}$$

Both terms are then equal to 87. The following code verifies the solution

codes/ap_equal.c

1.2.6 On a straight line passing through the foot of a tower, two points C and D are at distances of 4 m and 16 m from the foot respectively. If the angles of elevation from C and D of the top of the tower are complementary, then find the height of the tower. (CBSE 2017)

Solution: Let h be the height of the tower and θ be the angle of elevation from $\mathbf{C}$. The angle of elevation from $\mathbf{D}$ is then $90^\circ - \theta$. From (1.1.3.1),

$$\tan\theta = \frac{h}{4}, \quad \tan\left(90^\circ - \theta\right) = \frac{h}{16} \tag{1.2.6.1}$$

From (1.1.4.1),

$$\frac{h}{4} \cdot \frac{h}{16} = 1 \tag{1.2.6.2}$$

$$\implies h^2 = 64 \tag{1.2.6.3}$$

$$\therefore h = 8 \text{ m} \tag{1.2.6.4}$$

The following code generates Fig. 1.2.6.1

codes/trig_height.py

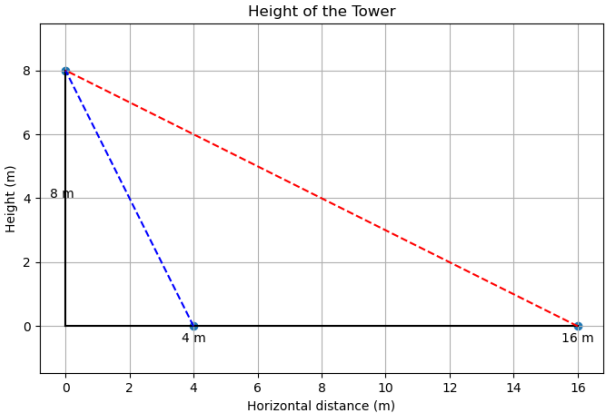


Fig. 1.2.6.1: Tower and the angles of elevation

1.2.7 Three semicircles each of diameter 3 cm, a circle of diameter 4.5 cm and a semicircle of radius 4.5 cm are drawn in the given figure. Find the area of the shaded region. (CBSE 2017)

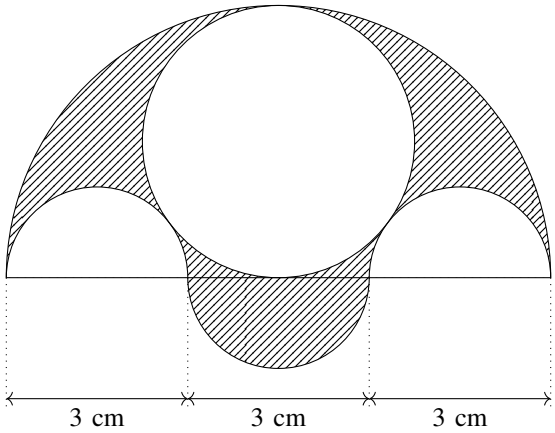


Fig. 1.2.7.1

Solution: In Fig. 1.2.7.1, the radii are

$$R = \frac{9}{2} \text{ cm (large semicircle)}, \quad r = \frac{3}{2} \text{ cm (small semicircles)}, \quad \rho = \frac{9}{4} \text{ cm (circle)} \tag{1.2.7.1}$$

The shaded region is the large semicircle, excluding the circle and the two upper small semicircles, and including the lower small semicircle. Using (1.1.8.1), the area

is

$$A = \frac{1}{2}\pi R^2 - \pi\rho^2 - 2\left(\frac{1}{2}\pi r^2\right) + \frac{1}{2}\pi r^2 \tag{1.2.7.2}$$

$$= \frac{1}{2}\pi R^2 - \pi\rho^2 - \frac{1}{2}\pi r^2 \tag{1.2.7.3}$$

$$= \pi\left(\frac{81}{8} - \frac{81}{16} - \frac{9}{8}\right) = \frac{63\pi}{16} \tag{1.2.7.4}$$

Substituting $\pi = \frac{22}{7}$,

$$A = \frac{63}{16} \times \frac{22}{7} = \frac{99}{8} = 12.375 \text{ cm}^2 \tag{1.2.7.5}$$

The following code generates Fig. 1.2.7.2

codes/shaded_area.py

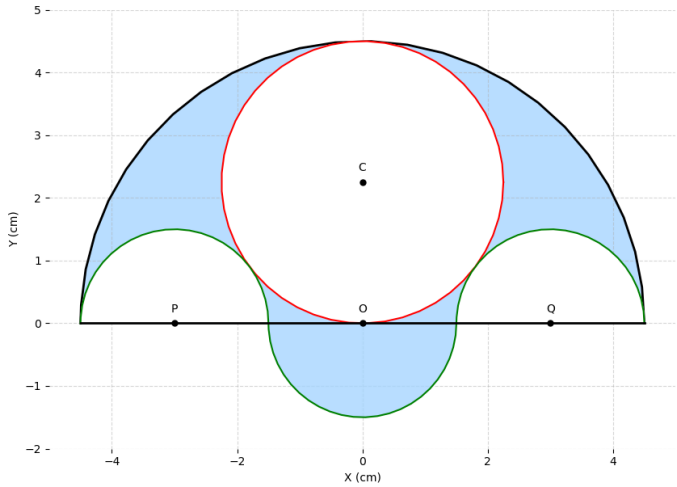


Fig. 1.2.7.2: Shaded region

1.2.8 In what ratio does the point $\left(\frac{24}{11}, y\right)$ divide the line segment joining the points $\mathbf{P}\,(2,-2)$ and $\mathbf{Q}\,(3,7)$? Also find the value of y. (CBSE 2017)

Solution: Let the ratio be $k : 1$. From (1.1.9.1),

$$\begin{pmatrix} \frac{24}{11} \\ y \end{pmatrix} = \frac{k\begin{pmatrix} 3 \\ 7 \end{pmatrix} + \begin{pmatrix} 2 \\ -2 \end{pmatrix}}{k+1} \tag{1.2.8.1}$$

Considering the first row,

$$\frac{24}{11} = \frac{3k+2}{k+1} \tag{1.2.8.2}$$

$$\implies 24k + 24 = 33k + 22 \tag{1.2.8.3}$$

$$\implies k = \frac{2}{9} \tag{1.2.8.4}$$

Thus, the ratio is $2:9$. Considering the second row,

$$y = \frac{7k-2}{k+1} = \frac{\frac{14}{9} - 2}{\frac{11}{9}} = -\frac{4}{11} \tag{1.2.8.5}$$

The following code verifies the solution

codes/section.c

1.2.9 Water in a canal, 5.4 m wide and 1.8 m deep, is flowing with a speed of 25 km/hour. How much area can it irrigate in 40 minutes, if 10 cm of standing water is required for irrigation ? (CBSE 2017)

Solution: The cross-sectional area, speed and time are

$$A = 5.4 \times 1.8 \text{ m}^2, \quad v = 25000 \text{ m/h}, \quad t = \frac{40}{60} = \frac{2}{3} \text{ h} \tag{1.2.9.1}$$

From (1.1.13.1), the volume of water that flows in 40 minutes is

$$V = 5.4 \times 1.8 \times 25000 \times \frac{2}{3} = 162000 \text{ m}^3 \tag{1.2.9.2}$$

If this water stands at a depth of 10 cm $= 0.1$ m over an area S,

$$S \times 0.1 = 162000 \tag{1.2.9.3}$$

$$\implies S = 1620000 \text{ m}^2 = 162 \text{ hectares} \tag{1.2.9.4}$$

1.2.10 In the given figure, two concentric circles with centre O have radii 21 cm and 42 cm. If $\angle AOB = 60°$, find the area of the shaded region. [Use $\pi = \frac{22}{7}$] (CBSE 2017)

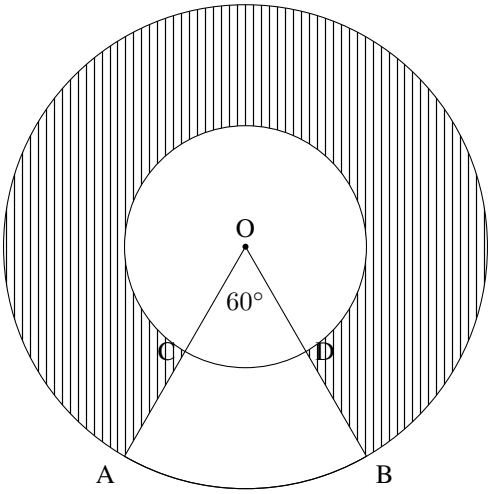


Fig. 1.2.10.1

Solution: In Fig. 1.2.10.1, the radii are $R = 42$ cm and $r = 21$ cm. The shaded region is the ring between the circles, excluding the part of the ring inside the $60°$ angle. From (1.1.15.1),

$$A = \pi\left(R^2 - r^2\right) - \frac{60°}{360°}\pi\left(R^2 - r^2\right) \tag{1.2.10.1}$$

$$= \frac{5}{6}\pi\left(42^2 - 21^2\right) = \frac{5}{6} \times \frac{22}{7} \times 1323 \tag{1.2.10.2}$$

yielding

$$A = 3465 \text{ cm}^2 \tag{1.2.10.3}$$

The following code generates Fig. 1.2.10.2

codes/sector_area.py

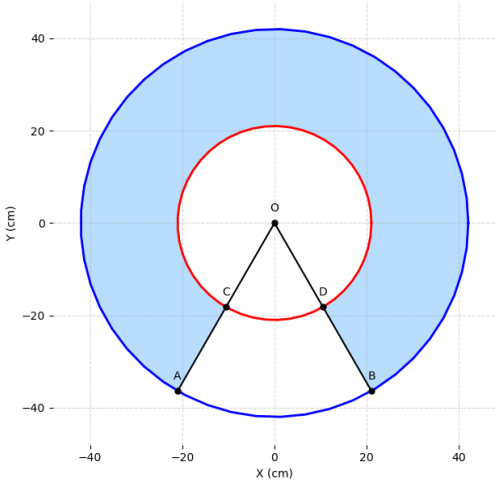


Fig. 1.2.10.2: Annular sector

1.2.11 The dimensions of a solid iron cuboid are $4.4\text{ m} \times 2.6\text{ m} \times 1.0\text{ m}$. It is melted and recast into a hollow cylindrical pipe of 30 cm inner radius and thickness 5 cm. Find the length of the pipe. (CBSE 2017)

Solution: The volume of the cuboid is

$$V = 4.4 \times 2.6 \times 1.0 = 11.44\text{ m}^3 \tag{1.2.11.1}$$

The inner and outer radii of the pipe are $r = 0.3$ m and $R = 0.3 + 0.05 = 0.35$ m. Let h be the length of the pipe. Equating the volumes using (1.1.14.1),

$$\frac{22}{7}\left(0.35^2 - 0.3^2\right)h = 11.44 \tag{1.2.11.2}$$

$$\implies \frac{22}{7} \times 0.0325\,h = 11.44 \tag{1.2.11.3}$$

$$\therefore h = 112\text{ m} \tag{1.2.11.4}$$

The following code verifies the solution

codes/cylinder.c

1.2.12 A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius on its circular face. The total height of the toy is 15.5 cm. Find the total surface area of the toy. (CBSE 2017)

Solution: The radius is $r = 3.5$ cm. The height of the cone is

$$h = 15.5 - 3.5 = 12\text{ cm} \tag{1.2.12.1}$$

From (1.1.16.1), the slant height is

$$l = \sqrt{12^2 + 3.5^2} = \sqrt{156.25} = 12.5\text{ cm} \tag{1.2.12.2}$$

The total surface area is the sum of the curved surface areas of the cone and the hemisphere, from (1.1.16.1) and (1.1.16.2),

$$A = \pi r l + 2\pi r^2 = \pi r \left(l + 2r\right) \tag{1.2.12.3}$$

$$= \frac{22}{7} \times 3.5 \times (12.5 + 7) = 214.5 \text{ cm}^2 \tag{1.2.12.4}$$

The following code generates Fig. 1.2.12.1

codes/cone_hemisphere.py

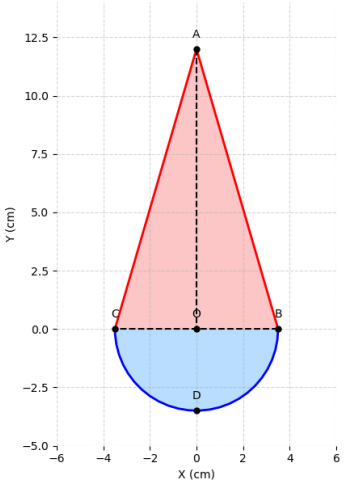


Fig. 1.2.12.1: Cone mounted on a hemisphere

1.2.13 How many terms of an A.P. $9, 17, 25, \ldots$ must be taken to give a sum of 636 ? (CBSE 2017)

Solution: Here $a = 9$ and $d = 8$. From (1.1.12.1),

$$\frac{n}{2}\left[18 + 8\left(n - 1\right)\right] = 636 \tag{1.2.13.1}$$

$$\implies 4n^2 + 5n - 636 = 0 \tag{1.2.13.2}$$

From (1.1.11.1),

$$n = \frac{-5 \pm \sqrt{25 + 10176}}{8} = \frac{-5 \pm 101}{8} \tag{1.2.13.3}$$

Since n is positive,

$$n = 12 \tag{1.2.13.4}$$

The following code verifies the solution

codes/ap_sum.c

1.2.14 If the roots of the equation $\left(a^2+b^2\right)x^2-2\left(ac+bd\right)x+\left(c^2+d^2\right)=0$ are equal, prove that $\frac{a}{b}=\frac{c}{d}$. (CBSE 2017)

Solution: From (1.1.11.2),

$$4\left(ac+bd\right)^2-4\left(a^2+b^2\right)\left(c^2+d^2\right)=0 \tag{1.2.14.1}$$

$$\implies a^2c^2+2abcd+b^2d^2=a^2c^2+a^2d^2+b^2c^2+b^2d^2 \tag{1.2.14.2}$$

$$\implies a^2d^2-2abcd+b^2c^2=0 \tag{1.2.14.3}$$

$$\implies \left(ad-bc\right)^2=0 \tag{1.2.14.4}$$

$$\implies ad=bc \tag{1.2.14.5}$$

Dividing by bd,

$$\frac{a}{b}=\frac{c}{d} \tag{1.2.14.6}$$

1.2.15 If the points $\mathbf{A}\left(k+1,2k\right)$, $\mathbf{B}\left(3k,2k+3\right)$ and $\mathbf{C}\left(5k-1,5k\right)$ are collinear, then find the value of k. (CBSE 2017)

Solution: Let

$$\mathbf{B}-\mathbf{A}=\begin{pmatrix}2k-1\\3\end{pmatrix},\quad \mathbf{C}-\mathbf{A}=\begin{pmatrix}4k-2\\3k\end{pmatrix} \tag{1.2.15.1}$$

From (1.1.10.1), the points are collinear if

$$\left|\begin{pmatrix}2k-1 & 4k-2\\3 & 3k\end{pmatrix}\right|=0 \tag{1.2.15.2}$$

$$\implies 3k\left(2k-1\right)-3\left(4k-2\right)=0 \tag{1.2.15.3}$$

$$\implies 2k^2-5k+2=0 \tag{1.2.15.4}$$

$$\implies \left(2k-1\right)\left(k-2\right)=0 \tag{1.2.15.5}$$

Hence,

$$k=\frac{1}{2}\quad\text{or}\quad k=2 \tag{1.2.15.6}$$

1.2.16 Two different dice are thrown together. Find the probability that the numbers obtained have

a) even sum, and
b) even product.

(CBSE 2017)

Solution: The total number of equally likely outcomes is $N=36$.

a) The sum is even if both numbers are even or both are odd. There are $3\times3=9$ outcomes of each type. From (1.1.17.1),

$$\Pr\left(\text{even sum}\right)=\frac{9+9}{36}=\frac{1}{2} \tag{1.2.16.1}$$

b) The product is odd only if both numbers are odd, which gives $3\times3=9$ outcomes.

Thus,

$$\Pr\left(\text{odd product}\right) = \frac{9}{36} = \frac{1}{4} \tag{1.2.16.2}$$

Using the complement in (1.1.17.1),

$$\Pr\left(\text{even product}\right) = 1 - \frac{1}{4} = \frac{3}{4} \tag{1.2.16.3}$$

The following code verifies the solution

codes/dice_prob.c

1.2.17 In the given figure, XY and $X'Y'$ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C, is intersecting XY at A and $X'Y'$ at B. Prove that $\angle AOB = 90°$. (CBSE 2017)

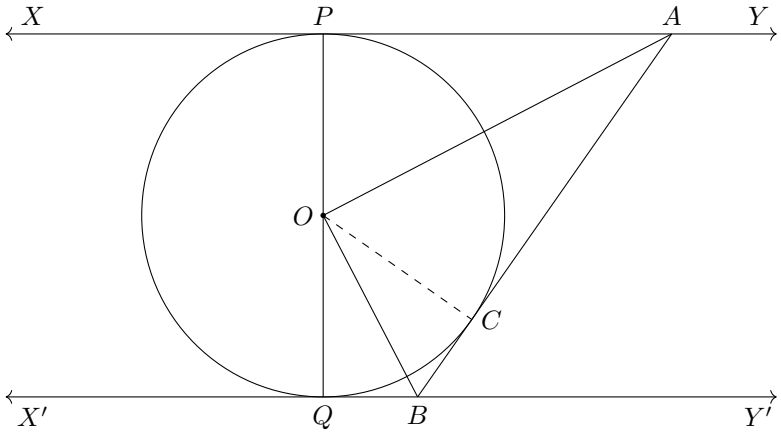


Fig. 1.2.17.1

Solution: In Fig. 1.2.17.1, let P and Q be the points of contact of XY and $X'Y'$. From (1.1.1.1), $OP \perp XY$ and $OQ \perp X'Y'$. Since $XY \parallel X'Y'$, the points P, O and Q are collinear. From (1.1.2.1),

$$AP = AC, \quad BQ = BC \tag{1.2.17.1}$$

Also $OP = OC = OQ$, and OA, OB are common. Therefore, the triangles OPA and OCA are congruent, and so are OQB and OCB,

$$\angle AOP = \angle AOC, \quad \angle BOQ = \angle BOC \tag{1.2.17.2}$$

Since POQ is a straight line,

$$\angle AOP + \angle AOC + \angle BOC + \angle BOQ = 180° \tag{1.2.17.3}$$

$$\implies 2\left(\angle AOC + \angle BOC\right) = 180° \tag{1.2.17.4}$$

Hence,

$$\angle AOB = \angle AOC + \angle BOC = 90^\circ \tag{1.2.17.5}$$

1.2.18 In a rain-water harvesting system, the rain-water from a roof of 22 m $\times$ 20 m drains into a cylindrical tank having diameter of base 2 m and height 3.5 m. If the tank is full, find the rainfall in cm. Write your views on water conservation. (CBSE 2017)

Solution: The radius of the tank is $r = 1$ m and its height is $h = 3.5$ m. From (1.1.18.1), the volume of the tank is

$$V = \frac{22}{7} \times 1^2 \times 3.5 = 11 \text{ m}^3 \tag{1.2.18.1}$$

Let the rainfall be x m. All the rain-water falling on the roof is collected in the tank, so

$$22 \times 20 \times x = 11 \tag{1.2.18.2}$$

$$\implies x = \frac{11}{440} = 0.025 \text{ m} \tag{1.2.18.3}$$

Hence, the rainfall is 2.5 cm. The following code verifies the solution

codes/rainfall.c

Water conservation is essential because fresh water is limited. Rain-water harvesting reduces the dependence on groundwater, recharges the water table and reduces flooding. Every individual should save water in daily life.

1.2.19 Prove that the lengths of two tangents drawn from an external point to a circle are equal. (CBSE 2017)

Solution: Let $\mathbf{P}$ be an external point of a circle with centre $\mathbf{O}$ and radius a. Let the tangents from $\mathbf{P}$ touch the circle at $\mathbf{T}_1$ and $\mathbf{T}_2$. From (1.1.1.1), the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are right angled at $\mathbf{T}_1$ and $\mathbf{T}_2$. By Pythagoras' theorem,

$$\|\mathbf{P} - \mathbf{T}_1\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - \|\mathbf{T}_1 - \mathbf{O}\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - a^2 \tag{1.2.19.1}$$

$$\|\mathbf{P} - \mathbf{T}_2\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - \|\mathbf{T}_2 - \mathbf{O}\|^2 = \|\mathbf{P} - \mathbf{O}\|^2 - a^2 \tag{1.2.19.2}$$

Therefore,

$$\|\mathbf{P} - \mathbf{T}_1\| = \|\mathbf{P} - \mathbf{T}_2\| \tag{1.2.19.3}$$

1.2.20 If the ratio of the sum of the first n terms of two A.Ps is $(7n + 1) : (4n + 27)$, then find the ratio of their 9th terms. (CBSE 2017)

Solution: Let S_n and S'_n be the sums of the first n terms of the two A.Ps, and a_9, a'_9 their 9th terms. Then

$$\frac{S_n}{S'_n} = \frac{7n + 1}{4n + 27} \tag{1.2.20.1}$$

For $m = 9$, $2m - 1 = 17$. From (1.1.19.1),

$$\frac{a_9}{a_9'} = \frac{S_{17}/17}{S_{17}'/17} = \frac{S_{17}}{S_{17}'} = \frac{7(17)+1}{4(17)+27} = \frac{120}{95} = \frac{24}{19} \tag{1.2.20.2}$$

Thus, the ratio of the 9th terms is $24 : 19$.

1.2.21 Solve for x :

$$\frac{x-1}{2x+1} + \frac{2x+1}{x-1} = 2, \quad \text{where } x \neq -\frac{1}{2}, 1$$

(CBSE 2017)

Solution: Multiplying by $(2x+1)(x-1)$,

$$(x-1)^2 + (2x+1)^2 = 2(2x+1)(x-1) \tag{1.2.21.1}$$
$$\implies 5x^2 + 2x + 2 = 4x^2 - 2x - 2 \tag{1.2.21.2}$$
$$\implies x^2 + 4x + 4 = 0 \tag{1.2.21.3}$$
$$\implies (x+2)^2 = 0 \tag{1.2.21.4}$$

Here $q^2 - 4pr = 16 - 16 = 0$, so from (1.1.11.2) the roots are equal,

$$x = -2 \tag{1.2.21.5}$$

which satisfies $x \neq -\frac{1}{2}, 1$.

1.2.22 A takes 6 days less than B to do a work. If both A and B working together can do it in 4 days, how many days will B take to finish it ? (CBSE 2017)

Solution: Let B take x days to finish the work. Then A takes $x - 6$ days. From (1.1.20.1),

$$\frac{1}{x} + \frac{1}{x-6} = \frac{1}{4} \tag{1.2.22.1}$$
$$\implies 4(2x-6) = x(x-6) \tag{1.2.22.2}$$
$$\implies x^2 - 14x + 24 = 0 \tag{1.2.22.3}$$

From (1.1.11.1),

$$x = \frac{14 \pm \sqrt{196-96}}{2} = \frac{14 \pm 10}{2} = 12 \text{ or } 2 \tag{1.2.22.4}$$

For $x = 2$, A would take -4 days, which is not possible. Therefore, B takes 12 days. The following code verifies the solution

codes/work_time.c

1.2.23 From the top of a tower, 100 m high, a man observes two cars on the opposite sides of the tower and in same straight line with its base, with angles of depression $30°$ and $45°$. Find the distance between the cars. [Take $\sqrt{3} = 1.732$] (CBSE 2017)

Solution: Let d_1 and d_2 be the distances of the cars from the base of the tower, for

the angles of depression 45° and 30° respectively. From (1.1.21.1),

$$\tan 45^\circ = \frac{100}{d_1} \implies d_1 = 100 \text{ m} \tag{1.2.23.1}$$

$$\tan 30^\circ = \frac{100}{d_2} \implies d_2 = 100\sqrt{3} \text{ m} \tag{1.2.23.2}$$

Since the cars are on opposite sides of the tower, the distance between them is

$$d_1 + d_2 = 100\left(1 + \sqrt{3}\right) = 100\left(1 + 1.732\right) = 273.2 \text{ m} \tag{1.2.23.3}$$

The following code generates Fig. 1.2.23.1

codes/tower_cars.py

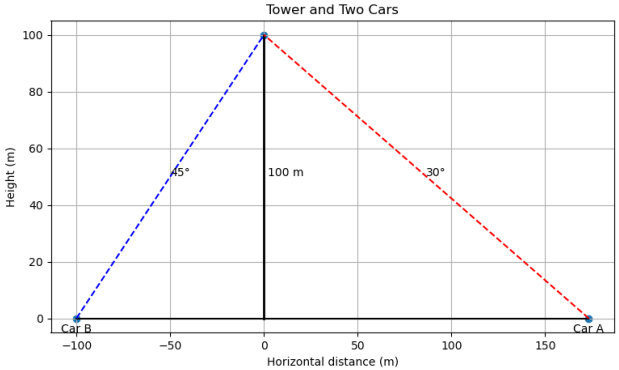


Fig. 1.2.23.1: Tower and the two cars

1.2.24 In the given figure, O is the centre of the circle with $AC = 24$ cm, $AB = 7$ cm and $\angle BOD = 90^\circ$. Find the area of the shaded region. (CBSE 2017)

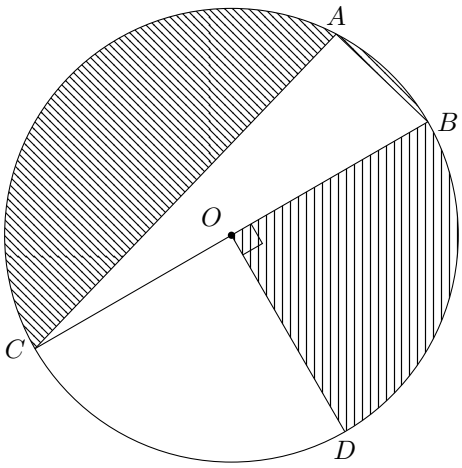


Fig. 1.2.24.1

Solution: In Fig. 1.2.24.1, B, O and C are collinear, so BC is a diameter. From (1.1.22.1),

$$\angle BAC = 90^\circ, \quad BC = \sqrt{7^2 + 24^2} = 25 \text{ cm} \tag{1.2.24.1}$$

The radius is $r = \frac{25}{2}$ cm. Since $OD \perp BC$, $\angle COD = 90^\circ$ as well. The circle consists of the triangle ABC, the two segments on AB and AC, and the two quadrants BOD and COD. The shaded region is the two segments and the quadrant BOD, so its area is

$$A = \pi r^2 - \text{ar}\,(\triangle ABC) - \text{ar}\,(\text{sector } COD) \tag{1.2.24.2}$$

$$= \pi r^2 - \frac{1}{2}\,(7)\,(24) - \frac{1}{4}\pi r^2 = \frac{3}{4}\pi r^2 - 84 \tag{1.2.24.3}$$

Substituting $\pi = \frac{22}{7}$ and $r^2 = \frac{625}{4}$,

$$A = \frac{3}{4} \times \frac{22}{7} \times \frac{625}{4} - 84 = \frac{20625}{56} - 84 = \frac{15921}{56} \approx 284.3 \text{ cm}^2 \tag{1.2.24.4}$$

The following code generates Fig. 1.2.24.2

codes/shaded_circle.py

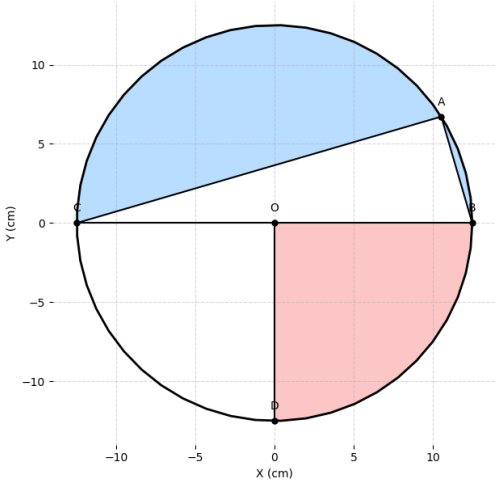


Fig. 1.2.24.2: Shaded region of the circle